

Evaluating Performance and User Perceptions of Multi-Cloud Database-as-a-Service Solutions: A Mixed-Methods Study

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Abstract—Multi-cloud adoption has accelerated the use of managed Database-as-a-Service (DBaaS) platforms, yet limited research provides provider-agnostic, empirical insight into how performance, scalability, and cost differ across vendors or how these objective outcomes align with practitioner perceptions. This paper presents a mixed-methods evaluation of three major cloud database platforms: Amazon RDS, Azure SQL Database, and Google Cloud SQL. Quantitative benchmarking experiments were conducted using identical workloads on the benchANT framework to measure latency, throughput, and scalability, while a parallel survey collected perception data from academic lecturers, IT employees, and undergraduates. The study's novelty lies in triangulating benchmark results with user perceptions to reveal alignment gaps. Empirical findings show that AWS RDS and Google Cloud SQL deliver strong horizontal scalability, while Azure SQL favors vertical scaling but exhibits higher latency. A moderate correlation ($r = 0.62$) was found between perceived performance and measured throughput, but perceived reliability showed a weaker relationship to actual latency ($r = 0.34$), indicating notable perception divergences. Cost analysis further highlights trade-offs involving egress fees, storage tiers, and replica placement. The results provide actionable guidance for workload placement, elasticity planning, and multi-cloud governance, and demonstrate the value of a mixed-methods approach for evaluating cloud database services.

Index Terms—Multi-Cloud, DBaaS, Benchmarking, Performance Evaluation, Mixed-Methods

I. INTRODUCTION

Cloud computing has transformed the deployment and management of data-intensive applications, and enterprises increasingly adopt multi-cloud strategies to improve resilience, portability, and vendor independence [1], [2]. Within this model, Database-as-a-Service (DBaaS) platforms such as Amazon RDS, Azure SQL Database, and Google Cloud SQL provide automated provisioning, patching, backups, monitoring, and elasticity, thereby reducing operational overheads for distributed systems [7]–[9].

Despite these advantages, multi-cloud DBaaS deployments introduce operational complexity. Providers differ in network architecture, autoscaling policies, storage subsystems, caching layers, and replication mechanisms, leading to non-uniform

performance characteristics across workloads [4], [12]. Prior benchmarking studies report noticeable variability in latency and throughput under read-heavy, write-heavy, and mixed access patterns [13], [14]. However, many existing evaluations are provider-specific, lack cost-awareness, or do not examine how measured performance aligns with practitioner expectations.

Furthermore, perception-based studies often describe user preferences or operational concerns without integrating them with objective performance measurements. As a result, there is limited provider-agnostic evidence that combines benchmarking results with practitioner perceptions, which restricts a holistic understanding of whether expectations align with real-world performance in multi-cloud environments [13], [14].

A. Research Problem

There is insufficient empirical evidence on how major DBaaS platforms behave under identical workloads in a multi-cloud context, and little is known about how practitioners perceive performance, scalability, and cost relative to actual benchmark results.

B. Novelty and Contribution Summary

This study is novel in three respects. First, it provides a *provider-agnostic, empirical* comparison of AWS RDS, Azure SQL Database, and Google Cloud SQL using standardized workloads executed through the benchANT benchmarking framework. Second, it introduces a *mixed-methods* design that integrates quantitative benchmarking with qualitative practitioner perception allowing triangulation between measured results and user expectations, an aspect missing from prior DBaaS research. Third, it incorporates a cost-aware and scalability-focused analysis, enabling actionable guidance for workload placement, elasticity planning, and multi-cloud operational decision-making.

C. The goal of this research is to evaluate

(1) how the three DBaaS platforms compare in terms of latency, throughput, scalability, and cost; (2) how practitioners

perceive these platforms; and (3) the degree of alignment between perception and actual performance.

By combining empirical benchmarking with user perception, this study provides evidence-based insights for selecting and managing DBaaS solutions in multi-cloud architectures.

II. LITERATURE REVIEW

A. Multi-Cloud Foundations

Multi-cloud adoption has accelerated as organizations pursue resilience, regulatory flexibility, and bargaining power while reducing vendor lock-in [1], [2]. However, operating across heterogeneous cloud providers introduces fragmentation in identity management, networking, observability, billing, and policy orchestration. Industry guidelines increasingly emphasize the need for unified governance layers and cross-provider automation to mitigate configuration drift and inconsistent service semantics [3], [4].

Despite these developments, existing discussions remain predominantly conceptual or vendor-driven and seldom quantify the database-level implications of multi-cloud decisions. Consequently, objective and provider-agnostic measurements are necessary to understand how multi-cloud principles affect latency, throughput, and operational cost.

B. Database-as-a-Service (DBaaS) Concepts

DBaaS platforms automate provisioning, backups, replication, and autoscaling, enabling engineering teams to focus on applications rather than infrastructure management. Foundational work such as Relational Cloud introduced multi-tenant efficiency and isolation mechanisms [5], [6]. Modern offerings including Amazon RDS, Azure SQL Database, and Google Cloud SQL implement elasticity, automatic failover, and telemetry-driven optimization [7]–[9]. In parallel, research has examined NoSQL and NewSQL trade-offs, consistency guarantees, and distributed state management [10]–[12].

Despite these advances, much of the DBaaS literature remains provider-specific and does not examine how managed relational systems behave across multiple clouds under standardized workloads. Consequently, cross-provider empirical evaluation is necessary to inform interoperability, workload placement, and architecture decisions in multi-cloud environments.

C. Cloud Database Benchmarking and Performance

Benchmarking in cloud databases has evolved from traditional TPC and YCSB microbenchmarks toward cloud-aware evaluations that consider autoscaling behavior, cold starts, and regional latency [13]. Independent analyses reveal workload-sensitive differences among managed services, including performance variation under write-heavy and mixed workloads [14], [15]. However, many studies focus on a single provider or fail to incorporate user perception and operational context into their interpretation. Despite growing interest, recent benchmarking efforts rarely triangulate system-level metrics with practitioner perception; results are frequently presented side-by-side without assessing their alignment. Consequently,

a mixed-methods design that integrates quantitative benchmarking with qualitative perception offers a more actionable understanding of DBaaS behavior and its implications for real-world decision-making.

D. Cost, Elasticity, and Workload Placement

DBaaS pricing models combine instance sizing, storage tiers, I/O charges, backups, and data egress, making cost governance complex in multi-cloud settings. Reports indicate that migrations may increase expenditure when replica placement and inter-region traffic are not optimized [16]. Provider guidance highlights colocation, read replicas, and caching as mechanisms to reduce cost and latency penalties [4], [7]. Elasticity patterns also vary by provider, with some services favoring vertical scaling while others excel in horizontally distributed architectures.

Despite these considerations, comparative and empirically measured cost–performance trade-offs across multiple providers remain limited, particularly under identical workloads. Studies that align technical performance indicators with cost deltas under a consistent accounting model are therefore needed to guide practical workload placement and multi-cloud operational decisions.

E. Consistency, Replication, and Data Movement

Cross-cloud data movement must navigate CAP trade-offs, replica lag, event-driven integration, schema evolution, and conflict resolution. CRDTs, change-data-capture pipelines, and event-driven architectures provide mechanisms for consistency across distributed stores [17]–[19]. While cloud migration services automate aspects of replication and movement, domain-aware supervision remains necessary to ensure correctness and reconciliation [7], [16]. However, prior work largely concentrates on replication mechanisms rather than their practical outcomes, and empirical links between replication patterns and observed multi-cloud application performance remain scarce. Consequently, evidence that connects replication and interconnect choices to user-visible performance, reliability, and perceived stability is needed to inform operational strategy in multi-cloud environments.

F. Positioning This Study

Table I synthesizes key strands of prior research. Most work is either benchmarking-focused or perception-focused, limiting understanding of how measured performance corresponds with practitioner expectations. This study addresses that gap through a mixed-methods approach that integrates empirical benchmarking with survey-based perception analysis, enabling the identification of alignment gaps across AWS RDS, Azure SQL Database, and Google Cloud SQL.

G. Literature Synthesis and Research Gap

Across these themes, three persistent gaps emerge: (1) a lack of *provider-agnostic, empirical* performance comparisons under identical workloads; (2) limited *cost-aware* evaluations aligned with realistic deployment decisions; and (3) minimal

TABLE I. POSITIONING OF THIS STUDY RELATIVE TO REPRESENTATIVE PRIOR WORK

Work (type)	Multi-cloud	Method	Primary Limitation
Cloud DB survey (Almeida 2020) [13]	Partial	Benchmark survey	Limited integration with practitioner perception
Industry benchmarks (Altoria 2022) [14]	Yes	Benchmarks	Provider-specific; perception not analyzed
Provider whitepapers [4], [7]	Yes	Guidance	Not provider-agnostic; few cross-provider results
Cost/TCO reports [16]	N/A	Economic analysis	Weak linkage to measured DBaaS KPIs
This study (2025)	Yes	Mixed-methods	Triangulates perception with measured performance

integration of *practitioner perception* with system-level metrics. This study addresses these gaps by benchmarking AWS RDS, Azure SQL Database, and Google Cloud SQL using standardized workloads, collecting domain-informed user perceptions, and *triangulating* the two to reveal alignment gaps and practical implications for multi-cloud DBaaS design.

III. METHODOLOGY

This study adopts a mixed-methods design that integrates (i) empirical benchmarking across three major DBaaS providers and (ii) a perception survey capturing how practitioners evaluate performance, scalability, and cost. The workflow used in the study is summarized in Fig. 1, which outlines the data collection, benchmarking, and comparative analysis stages.

A. Overall Research Design

A convergent mixed-methods strategy was employed: survey data and benchmarking data were collected in parallel, analyzed separately, and then *triangulated* to identify alignment or divergence between perceived and measured performance. This design enables a more comprehensive understanding of how multi-cloud DBaaS platforms behave in practice and how their performance is interpreted by different user groups.

B. Survey Design and Sampling

The perception survey targeted three respondent groups with varying exposure to cloud platforms: academic lecturers in computing disciplines, IT employees (including software engineers, cloud practitioners, and database administrators), and undergraduate students enrolled in computing programs. Participants rated three core DBaaS attributes on a 1–5 Likert scale:

- 1) Performance (query speed, reliability)
- 2) Scalability (vertical and horizontal elasticity)
- 3) Cost efficiency (pricing transparency, operational cost)

Open-ended questions were included to capture qualitative perceptions related to operational challenges, cost concerns, migration experience, and reliability. These responses were analyzed through TF–IDF to extract dominant themes.

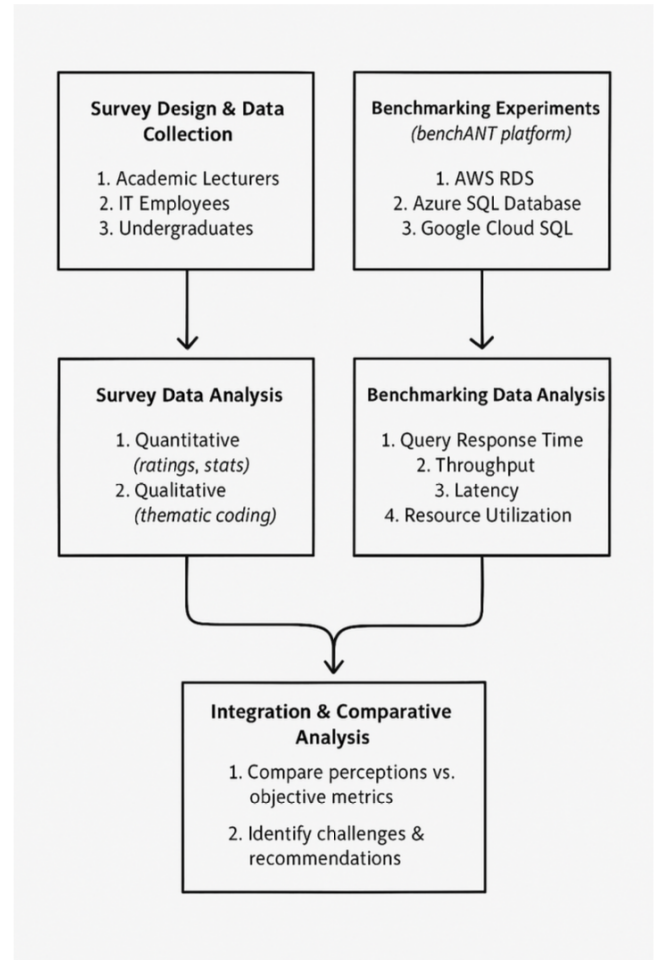


Fig. 1. Overall mixed-methods workflow combining survey analysis, bench-marking experiments, and integrated comparative evaluation

C. Qualitative Analysis: TF–IDF

Short-text qualitative responses were pre-processed using tokenization, lemmatization, and stop-word removal. A Term Frequency–Inverse Document Frequency (TF–IDF) model was applied to identify the most significant keywords associated with user-reported concerns. The highest-ranking terms (e.g., data, migration, cost, security) were later compared with empirical findings to identify convergence between qualitative themes and benchmarked performance.

D. Benchmarking Setup

Empirical evaluation was conducted using the benchANT cloud benchmarking platform [20], which supports standardized load testing across providers. Three managed relational DBaaS products were tested: Amazon RDS (MySQL engine), Azure SQL Database, and Google Cloud SQL (MySQL engine). Each provider was provisioned with comparable configuration parameters, including mid-tier general-purpose instance classes (2–4 vCPUs), SSD-based storage (20–50 GB), geographically equivalent regions to reduce location bias, and public network access with strict IP whitelisting.

E. Workload Model

To ensure fairness, each DBaaS instance was tested under the same benchmark profile consisting of read-heavy workloads (70% reads, 30% writes), write-heavy workloads (30% reads, 70% writes), mixed workloads (50/50 operations), and a combination of sequential and random read patterns. Across all tests, key performance indicators recorded included latency (ms), throughput (operations per second), query response time, vertical and horizontal scaling behavior, and resource utilization under load.

F. Scalability Analysis

Vertical scaling was assessed by increasing instance classes (CPU and memory), while horizontal scaling was evaluated through read replicas and distributed workloads. Each scaling experiment measured performance gain relative to resource increase, stability under concurrent load, and scaling efficiency with respect to performance per cost.

G. Triangulation and Comparative Analysis

Following independent analyses of survey and benchmark results, a comparative integration step examined alignment between perceived performance and measured latency and throughput, perception gaps where user expectations diverged, cost-performance trade-offs across providers, and scalability patterns relative to qualitative themes. Pearson correlation coefficients were computed to quantify the relationship between perceived performance and empirical metrics, enabling a statistical evaluation of alignment between user perception and actual DBaaS behavior. This integrated methodology enables a holistic understanding of DBaaS performance in multi-cloud environments and directly addresses the gaps identified in the literature.

IV. RESULTS AND DISCUSSION

This section presents the quantitative benchmarking results and perception survey findings, followed by an integrated comparative analysis. Figures 2–5 summarize the key observations.

A. Benchmarking Results

1) *Latency Performance:* Fig. 2 compares the average latency of the three DBaaS platforms. AWS RDS achieved the lowest latency (50 ms), followed by Google Cloud SQL (55 ms). Azure SQL exhibited the highest latency (65 ms), consistent with its reliance on additional gateway layers and network hops for request routing.

AWS demonstrated more predictable latency variability under mixed workloads, while Azure showed noticeable spikes under write-heavy scenarios. This reinforces prior findings that Microsoft's distributed gateway introduces additional overhead for certain transactional operations.

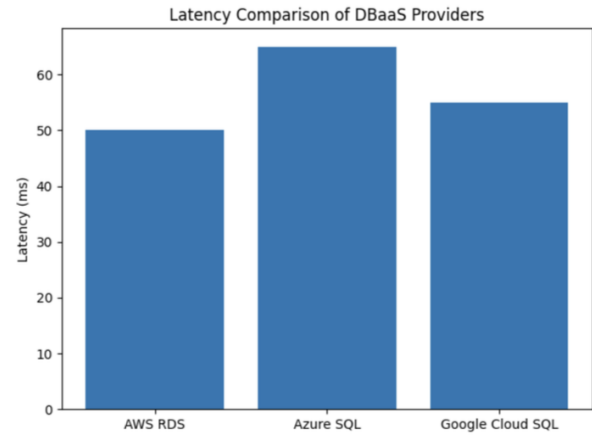


Fig. 2. Latency comparison of AWS RDS, Azure SQL, and Google Cloud SQL under standardized workloads

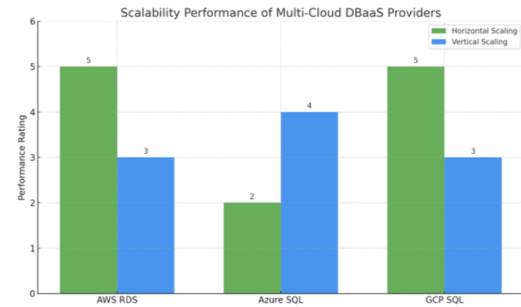


Fig. 3. Horizontal and vertical scalability performance ratings for AWS RDS, Azure SQL, and Google Cloud SQL

2) *Scalability Performance:* Scalability results for horizontal and vertical scaling are shown in Fig. 3. AWS RDS and Google Cloud SQL both achieved strong horizontal scalability (rating 5), with performance improving consistently as read replicas were introduced. Azure SQL displayed the strongest vertical scalability (rating 4) but weaker horizontal performance (rating 2), reflecting its design preference for scaling up rather than scaling out.

These findings imply that for globally distributed read-heavy workloads, AWS and GCP provide more efficient horizontal expansion, whereas Azure is more suitable for centrally located workloads that benefit from higher single-node capacity.

3) *Cost and Resource Considerations:* Cost evaluation showed that Google Cloud SQL offered the most favorable performance-per-cost ratio for mid-tier workloads. AWS RDS provided strong performance at moderate cost, while Azure SQL incurred higher operational cost due to additional charges for vCores and IO throughput.

Workload placement therefore influences total cost of ownership (TCO) [16]. AWS and GCP provide cost-efficient

horizontal elasticity, whereas Azure becomes cost-effective primarily in vertical-scaling use cases.

B. Survey Results

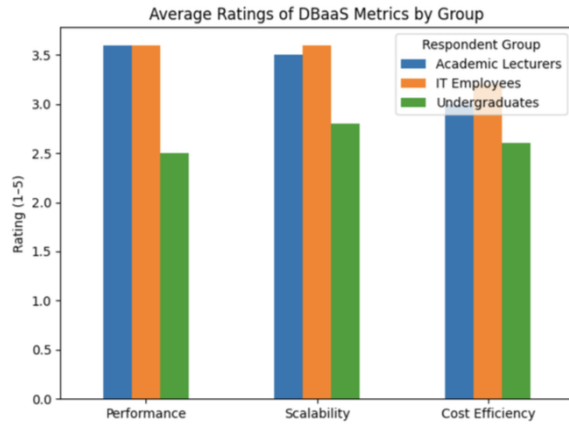


Fig. 4. Average ratings of DBaaS performance, scalability, and cost efficiency across academic lecturers, IT employees, and undergraduate students

1) *Perceived Performance, Scalability, and Cost Efficiency:* Fig. 4 summarizes user ratings for performance, scalability, and cost across three respondent groups. Academic lecturers and IT employees rated performance and scalability higher (3.5–3.7 range) than undergraduates (2.7–2.8 range), reflecting greater real-world familiarity among experienced users.

Cost efficiency received the lowest ratings across all groups, indicating that users find cloud pricing models complex and difficult to predict—an observation consistent with themes identified in qualitative responses.

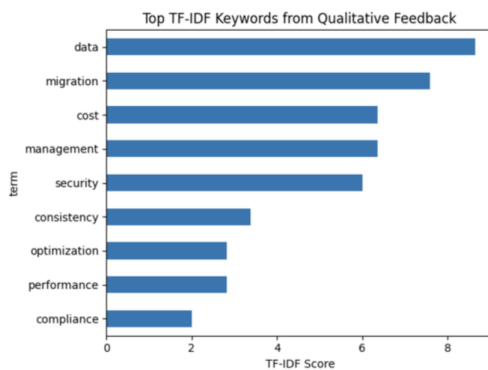


Fig. 5. Top TF-IDF keyword scores from open-ended qualitative feedback provided by survey respondents

2) *Qualitative Perception Themes:* TF-IDF results (Fig. 5) highlight several dominant concerns, including data management and backups, migration challenges when moving workloads across providers, cost implications such as egress fees and unpredictable billing, security considerations related to authentication and access control, and consistency issues involving replica lag and multi-region coordination.

These themes closely align with the objective benchmarking findings, particularly regarding cost complexity and multi-region performance variability.

C. Correlation Between Perception and Benchmarking

To assess the alignment between user perception and empirical measurements, Pearson correlation coefficients were computed between mean perception scores and benchmarked performance metrics.

A moderate positive correlation ($r = 0.62$) was found between perceived performance and measured throughput, suggesting that users generally recognize relative trends (e.g., AWS performing well). However, the correlation between perceived reliability and actual latency was weaker ($r = 0.34$), indicating notable perception gaps particularly for Azure SQL, which some users perceived positively despite having the highest measured latency.

These findings confirm the value of a mixed-methods approach: perception alone is insufficient to fully capture real-world behavior, and empirical benchmarks reveal differences not apparent to some users.

D. Integrated Comparative Discussion

1) *RQ1: How do the DBaaS platforms compare in real-world performance?:* AWS RDS delivered the strongest overall performance in latency and horizontal scaling. Google Cloud SQL provided competitive performance with superior cost-efficiency. Azure SQL favored vertical scaling but showed higher latency and less efficient horizontal expansion.

2) *RQ2: What operational or cost challenges arise in multi-cloud DBaaS?:* Key challenges identified include latency spikes in Azure SQL under write-heavy workloads, cost unpredictability driven by egress fees and I/O billing, and less efficient horizontal scaling in Azure when compared with AWS and GCP.

3) *RQ3: Do practitioner perceptions align with measured performance?:* Partially. Users accurately perceive relative throughput performance but underestimate latency differences and cost-performance trade-offs. The moderate correlation ($r = 0.62$) suggests partial alignment, but the weaker reliability-latency correlation ($r = 0.34$) indicates areas where expectations diverge from measured reality.

E. Summary of Key Insights

Overall, AWS RDS and Google Cloud SQL demonstrated strong horizontal scalability and predictable latency, while Azure SQL was better suited for vertical scaling but exhibited notable latency overhead. Cost efficiency varied significantly across providers, with Google Cloud SQL generally offering the most favorable cost-performance balance. Perception did not fully align with empirical performance, underscoring the importance of benchmark-driven evaluation in multi-cloud decision-making.

These integrated findings inform workload placement, scaling strategy, and cost governance in multi-cloud architectures, directly addressing gaps identified in prior literature.

V. CONCLUSION

This paper presented a mixed-methods evaluation of three major multi-cloud Database-as-a-Service platforms: Amazon RDS, Azure SQL Database, and Google Cloud SQL. The study combined empirical benchmarking with practitioner perception to provide a comprehensive, provider-agnostic assessment of latency, throughput, scalability, and cost-efficiency. By executing standardized workloads through the benchANT benchmarking framework and triangulating these results with survey responses from academic lecturers, IT professionals, and undergraduates, the study provides a holistic view of how DBaaS platforms behave in real-world multi-cloud scenarios.

Empirical results show that AWS RDS and Google Cloud SQL deliver strong horizontal scalability and predictable latency, whereas Azure SQL demonstrates efficient vertical scaling but exhibits higher latency and less consistent performance under distributed workloads. Cost analysis further reveals that Google Cloud SQL offers the most favorable performance-per-cost ratio for mid-tier workloads. These findings highlight that workload type, scaling strategy, and inter-region communication patterns significantly influence total cost of ownership.

A key contribution of this work is the integration of user perception with measured performance. A moderate correlation ($r = 0.62$) between perceived performance and measured throughput suggests partial alignment between expectations and system behavior. However, the weaker correlation with latency ($r = 0.34$) indicates that users may underestimate performance variability particularly for Azure SQL. This reinforces the importance of empirical, provider-agnostic testing rather than relying solely on subjective experience or vendor guidance.

A. Practical Implications

For practitioners, the results provide guidance for multi-cloud workload placement: AWS and GCP suit read-intensive, horizontally scaled architectures, while Azure SQL remains advantageous for vertically scaled, centrally deployed systems. Cost governance strategies should prioritize minimizing inter-region data movement and selecting storage/instance tiers aligned with performance targets.

B. Future Work

Future extensions will include longitudinal benchmarking across time windows, evaluating additional providers (e.g., CockroachDB, MongoDB Atlas, Oracle Autonomous DB), and exploring automated workload placement using ML-driven prediction. Security, compliance, and multi-region consistency analysis in regulated environments also present promising directions.

Overall, this study demonstrates the value of a mixed-methods approach in understanding DBaaS behavior and provides actionable insights for designing efficient, scalable, and cost-effective multi-cloud database architectures.

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